我的提问：

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865123482_...12_L.mp4FT

CSV

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data_feature

PY

The provided CSV file contains data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb to perform continuous finger tapping. A 10-second segment from this activity was selected for analysis. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement, employing a tracking technique to record the x and y coordinates of the stickers on the thumb and index finger in each frame throughout the 10-second interval.

The objective is to score the subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING based on the selected 10 seconds of continuous finger tapping. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

“Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

0: Normal: No problems.

1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.

2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.

3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.

4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

The provided Python file includes a method for calculating the number of taps. Use this method for the analysis and begin by generating a plot of the data.

To determine the subject's score, analyze the provided data thoroughly and give a detailed motivation for the assigned score based on the MDS-UPDRS finger tapping instruction. Print the score using only one number on the last line of your answer, after the detailed motivation.

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顯示程式碼

Python

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.signal import find_peaks
import numpy as np
```

```
# Define the file path
data_path = "865123482_20200713_Hands_R_12_L.mp4FT.csv"

try:
    # Load the data
    data = pd.read_csv(data_path)

    # --- Begin logic adapted from data_feature.py ---

    # Calculate the Euclidean distance between thumb and index finger
    data['Distance'] = np.sqrt((data['X_thumb'] - data['X_index'])**2 + (data['Y_thumb'] - data['Y_index'])**2)

    # Check for NaNs in Distance (e.g., from missing data)
    if data['Distance'].isnull().any():
        print("Warning: NaNs found in Distance calculation. Dropping rows with NaNs.")
        data.dropna(subset=['Distance'], inplace=True)

    # Normalize the distance
    min_dist = np.min(data['Distance'])
    max_dist = np.max(data['Distance'])

    if max_dist == min_dist:
        print("Error: Max and Min distance are identical. Cannot normalize.")
        data['Distance_Norm'] = 0.5
    else:
        data['Distance_Norm'] = (data['Distance'] - min_dist) / (max_dist - min_dist)

    # Identify peaks (taps) in the normalized distance
    # Using the parameters from the provided .py file: height=0.2, distance=30
    peaks, _ = find_peaks(data['Distance_Norm'], height=0.2, distance=30)

    # Calculate metrics for scoring

    # 1. Tap Amplitudes (from the raw, non-normalized distance)
    tap_amplitudes = data['Distance'].iloc[peaks]

    # 2. Tap Periods (time between taps)
    tap_periods_frames = np.diff(data['Frame'].iloc[peaks])
    fps = 240.0  # Frames per second
    tap_periods_sec = tap_periods_frames / fps

    # 3. Number of Taps
    num_taps = len(peaks)

    # 4. Total Duration
    total_frames = data['Frame'].max() - data['Frame'].min()
    total_duration_sec = total_frames / fps

    # --- Print values for analysis ---
    print(f"--- FINGER TAPPING ANALYSIS METRICS ---")
    print(f"Data file: {data_path}")
    print(f"Total Frames Analyzed: {total_frames + 1}") # +1 to be inclusive
    print(f"Total Duration (seconds): {total_duration_sec:.2f}")
```

```
    print(f"Frames per Second (FPS): {fps}")
    print(f"\n--- Tapping Performance ---")
    print(f"Total Detected Taps: {num_taps}")

    print(f"\nTap Amplitudes (raw distance units, sequential):")
    print(np.array(tap_amplitudes))

    print(f"\nTap Periods (time between taps, in seconds):")
    print(np.array(tap_periods_sec))

    if len(tap_periods_sec) > 0:
        print(f"Mean Tap Period (seconds): {np.mean(tap_periods_sec):.4f}")
        print(f"Std Dev of Tap Period (rhythm): {np.std(tap_periods_sec):.4f}")
        print(f"Min Tap Period (fastest): {np.min(tap_periods_sec):.4f}")
        print(f"Max Tap Period (slowest): {np.max(tap_periods_sec):.4f}")

    # --- Plotting logic from data_feature.py ---
    plt.figure(figsize=(15, 6))
    plt.plot(data['Frame'], data['Distance_Norm'], label='Normalized Distance (Thumb-Index)')
    plt.plot(data['Frame'].iloc[peaks], data['Distance_Norm'].iloc[peaks], "x", markersize=8, label='Detected Taps
(Peaks)')
    plt.title('Finger Tapping Analysis: Normalized Distance vs. Frame')
    plt.xlabel('Frame Number')
    plt.ylabel('Normalized Distance')
    plt.legend()
    plt.grid(True)
    plt.savefig('finger_tap_analysis.png')
    plt.close()

    print("\nPlot 'finger_tap_analysis.png' successfully generated.")

except FileNotFoundError:
    print(f"Error: The file {data_path} was not found.")
except Exception as e:
    print(f"An error occurred: {e}")
```

程式碼輸出內容

```
--- FINGER TAPPING ANALYSIS METRICS ---
Data file: 865123482_20200713_Hands_R_12_L.mp4FT.csv
Total Frames Analyzed: 1637
Total Duration (seconds): 6.82
Frames per Second (FPS): 240.0

--- Tapping Performance ---
Total Detected Taps: 12

Tap Amplitudes (raw distance units, sequential):
[193.93297811 185.3968716  196.98730924 191.25375813 199.75234667
 202.95073294 208.59769893 212.86850401 205.54804791 216.08563117
 207.38611332 205.13897728]

Tap Periods (time between taps, in seconds):
```

```
[0.25       0.63333333 0.75416667 0.625      0.61666667 0.57916667
 0.60833333 0.675      0.5875     0.6        0.59583333]
Mean Tap Period (seconds): 0.5932
Std Dev of Tap Period (rhythm): 0.1183
Min Tap Period (fastest): 0.2500
Max Tap Period (slowest): 0.7542

Plot 'finger_tap_analysis.png' successfully generated.
```

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Here is the analysis of the finger-tapping performance based on the provided data and scoring criteria.

Finger Tapping Performance Analysis

The provided Python script was executed to analyze the 6.82-second (1637 frames) data segment. The script calculated the frame-by-frame distance between the thumb and index finger, normalized this distance, and then identified peaks in the movement, which correspond to the moments the fingers are farthest apart (i.e., one "tap" cycle).

Here is the plot of the normalized distance over time, with the 12 detected taps marked by an 'x':

Key Metrics and Scoring Motivation

The analysis yielded the following key metrics, which are used to determine the score based on the MDS-UPDRS item 3.4 criteria.

- **Total Taps Detected:** 12

- **Tap Amplitudes (Raw Distance):** `[193.9, 185.4, 197.0, 191.3, 199.8, 203.0, 208.6, 212.9, 205.5, 216.1, 207.4, 205.1]`

- **Tap Periods (Seconds between taps):** `[0.25, 0.63, 0.75, 0.63, 0.62, 0.58, 0.61, 0.68, 0.59, 0.60, 0.60]`

Based on these results, the score is motivated as follows:

1. **Amplitude (Size of Taps):** The scoring criteria penalize "decrementing amplitude." The analysis of the tap amplitudes shows **no decrement**. After an initial slight dip, the amplitude values are consistently maintained and even increase, peaking at the 10th tap (216.1). This indicates the subject successfully followed the instruction to tap "as big as possible" without fatiguing. This rules out scores 1(c), 2(c), and 3(c).

2. **Speed (Slowing):** The criteria also penalize "slowing." The sequence of tap periods does not show progressive slowing. After the initial irregular taps, the period stabilizes between 0.58 and 0.68 seconds. This performance is inconsistent with "slight slowing," "mild slowing," or "moderate slowing." This rules out scores 1(b), 2(b), and 3(b).

3. **Hesitations/Halts (Rhythm):** The criteria for Score 1 include: **"a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement."**

   - The tap periods data clearly supports this. The sequence begins with an abnormally fast tap (0.25s), followed by two much slower taps (0.63s and 0.75s). This 0.75s period, in particular, represents a clear hesitation or break in rhythm when compared to the subsequent, more stable tapping pace (around 0.60s).

- This pattern perfectly matches the description for Score 1, criterion (a). The performance is not "Normal" (Score 0) due to this arrhythmic start, but the impairment is limited to these "one or two interruptions" and does not involve the more severe issues of amplitude decrement or progressive slowing.

Conclusion: The subject maintains excellent amplitude and does not exhibit progressive slowing. The performance is downgraded from "Normal" only because of a clear break in rhythm at the beginning of the task.

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